

Homework 3. Random Walks

You can get up to 10 points for this homework.

Question 1. For the following tasks use Python and submit one file incorporating all your answers, named as "H3-YourFirstName-YourLastName.ipynb".

- ☐ A The historical daily closing price of many stocks is available for free online. Select a stock ticker of your choice, download its historical daily closing price for 1 year, create a .csv file out of those data and put it online (you will use a link to it in your program).
- ☐ B Plot the daily closing price of your chosen stock.
- ☐ C Determine if it is a random walk or not. If it is not a random walk, explain why.
- ☐ D Keep the last 5 days of data as a test set. The rest will be the train set.
- ☐ E Forecast the last 5 days using naive forecasting methods, and measure the MSE. Which method is the best?
- ☐ F Plot your forecasts.
- ☐ G Forecast the next timestep over the test set, and measure the MSE. Did it decrease?
- ☐ H Plot your forecasts.